Sparse Signal-Field SAEs for Hierarchical Vision Representations

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Abstract

We study whether intermediate vision activations are sparse in a learned transform domain and whether those sparse codes form a stable hierarchy across depth. Using frozen ResNet backbones on CIFAR-100 and CIFAR-10, we train sparse autoencoders (SAEs) on activation fields at five depths and retain only a small fraction of learned feature-location coefficients. A convolutional Field SAE preserves downstream classification at 2–5% retained coefficients across all sections, materially outperforming both a pointwise Vector SAE and linear or random-code baselines. We then test whether sparse codes at one depth predict sparse codes at the next. Single-step transitions are strong, and a frozen-block hybrid chain shows that the sparse support is information-sufficient end-to-end. Naive learned multi-step chaining initially collapses, but re-grounding, scheduled-sampling, and later rollout-aligned retraining recover most of the gap to the causal upper bound. Follow-up runs strengthen the story with seed stability, transfer to ResNet-110 and ResNet-20/CIFAR-10, sparse-vs-dense transition taxonomy, causal intervention evidence, and a first ViT transfer artifact. The resulting picture is that learned sparse field codes are both behavior-preserving and structurally meaningful, while stable hierarchical rollout depends on matching training and evaluation to the sparse-code manifold.

1 Introduction

Sparse autoencoders have become a useful way to probe learned representations, but most SAE work treats activations as vectors. Convolutional vision backbones instead expose activation fields with explicit spatial structure, so a sparse representation should preserve both *what* feature is active and *where* it is active. This project asks two questions:

1. Can a CNN activation field be reconstructed from a very sparse set of learned feature-location coefficients without changing the network's prediction?

2. Do sparse codes at one depth generate sparse codes at the next, forming a depth-wise sparsity tree?

Our core setting is a frozen ResNet-56 on CIFAR-100 with baseline top-1 accuracy of 71.83%. We tap five sections ($Q1 \dots Q5$), train SAEs independently at each section, and evaluate reconstructions by splicing them back into the frozen model and measuring the resulting classifier behavior. We then learn section-to-section transitions in sparse-code space and compare them against a frozen-block hybrid upper bound that uses the real backbone blocks between sparse codes.

The main contributions are:

- A convolutional *Field SAE* that learns sparse codes over feature-location fields and strongly outperforms a pointwise *Vector SAE* at aggressive budgets.
- Evidence that 2–5% retained sparse coefficients are sufficient to preserve near-original behavior across five CNN depths.
- A hierarchy study showing that sparse support is causally sufficient end-to-end, while stable learned chaining requires re-grounding or rollout-aligned training.
- Follow-up robustness and transfer results spanning seed stability, new backbones, sparse-vs-dense transition comparison, causal interventions, and an initial ViT artifact.

2 Method

2.1 Backbone, sections, and evaluation

We use a frozen ResNet-56 trained on CIFAR-100 and evaluate five post-activation sections:

Section	Shape	Description
$Q1$	$16 \times 32 \times 32$	post-stem
$Q2$	$16 \times 32 \times 32$	end of stage 1
$Q3$	$32 \times 16 \times 16$	end of stage 2
$Q4$	$64 \times 8 \times 8$	mid stage 3
$Q5$	$64 \times 8 \times 8$	end of stage 3

For each section we cache activations and train an SAE with overcomplete width $K = 8C$. Given activation field $H \in \mathbb{R}^{C\times H\times W}$, the SAE produces a code field $Z \in \mathbb{R}^{K\times H\times W}$, applies a hard TopK mask $\tilde{Z}$, and reconstructs $\hat{H}$. Reconstruction quality is measured by relative ℓ_2, cosine similarity, and most importantly *spliced top-1*: replace the true activation with $\hat{H}$, run the frozen tail, and compare classifier outputs to the original network.

2.2 SAE architectures and masks

We compare two SAE families:

- **Vector SAE:** a shared pointwise encoder/decoder applied independently at each spatial location.
- **Field SAE:** a convolutional encoder/decoder with local residual blocks and depthwise spatial mixing before the sparse bottleneck.

We also compare two masking schemes:

- **Global mask:** retain the top fraction of coefficients over the whole field.
- **RF-local mask:** retain top coefficients within receptive-field windows of the next section.

SAEs are trained with a staged curriculum: warmup without sparsity, annealing from 50% retained to the target budget, then fixed-budget training. This curriculum substantially stabilizes learning relative to a cold start at the final sparsity level.

2.3 Hierarchy experiments

For the winner SAE configuration we learn adjacent transitions $T_t : \tilde{Z}_t \to \hat{Z}_{t+1}$ and evaluate:

- **Single-step learned transitions:** sparse code at Q_t predicts sparse code at Q_{t+1}.
- **Frozen-block hybrid chain:** decode sparse code, run the real frozen backbone block, then re-encode; this isolates whether the sparse support contains enough information independent of predictor error.
- **Multi-step learned chaining:** compose all learned transitions from $Q1$ to $Q5$, optionally with re-masking or re-grounding.

Re-grounding means decode $\to$ re-encode $\to$ re-mask after each predicted step, projecting the rollout back onto the SAE manifold.

3 Sparse Reconstruction Results

Phase 1 establishes the anchor result: sparse reconstruction is enough to preserve downstream behavior.

Table 1 and Figure 1 show that every section reaches near-original top-1 accuracy with only 2–10% of coefficients retained, and deeper sections preserve accuracy even at 2%. The result is not trivial autoencoding: a random-dictionary control is near chance, while a matched-budget PCA baseline is consistently weaker.

This confirms the transform-domain sparsity hypothesis for the anchor backbone: hidden fields may look dense in pixel-channel coordinates, yet they are highly compressible in a learned sparse basis.

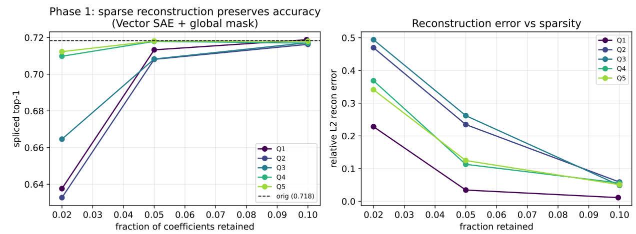

Figure 1: Phase 1 reconstruction curves. Spliced accuracy stays near the original classifier performance at small retained fractions, especially in deeper sections.

Table 1: Representative Vector SAE + global-mask results from Phase 1.

Section	Fraction	RelL2	Cos	Top-1
Q1	5%	0.034	0.999	0.713
Q2	10%	0.059	0.998	0.716
Q3	10%	0.049	0.999	0.717
Q4	2%	0.369	0.929	0.710
Q5	2%	0.341	0.939	0.712

4 Architecture and Masking Choices

Phase 2 compares the SAE architecture and mask type. The central result is that the convolutional Field SAE materially outperforms the pointwise Vector SAE at aggressive budgets, especially in early high-resolution sections.

The Field SAE winner path is therefore the main basis for later hierarchy experiments. The original RF-local masking story looked much weaker, but a later fair-budget rerun revised that interpretation: once RF budgets are normalized fairly per window, RF-local masking becomes competitive with the global path rather than fundamentally broken.

5 Sparse-Code Hierarchy

5.1 Single-step transitions

Adjacent sparse-code transitions are generally strong, especially on easier edges.

The learned predictor exactly matches the hybrid bound on $Q1 \to Q2$, while $Q3 \to Q4$ remains the hardest

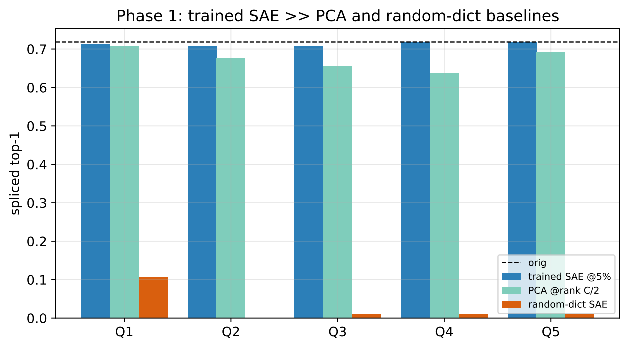

Figure 2: The trained SAE strongly outperforms PCA and a random-dictionary control. Sparse reconstruction quality is therefore learned rather than a budget artifact.

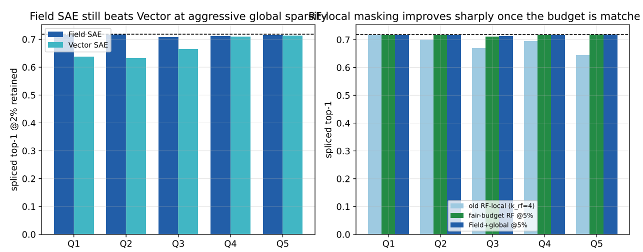

Figure 3: Field SAE versus Vector SAE. Local spatial mixing is especially valuable in early sections at low retained fractions.

step throughout the project. That persistent bottleneck later reappears in both ResNet-110 and taxonomy follow-ups, suggesting that it reflects a genuine representational difficulty rather than simple noise.

5.2 Multi-step chaining

The first raw learned chain collapses under recursive rollout even though the hybrid chain preserves near-original accuracy. This separation is crucial: the sparse support contains enough information, but naive sparse-code prediction suffers from exposure bias and manifold drift.

At first glance, these results suggested that re-grounding was strictly necessary. Later changed reruns refined that conclusion. A rollout-aligned *raw-only* retraining run, with scheduled sampling, full-chain BPTT, and gradient clipping but without re-grounding on the carrier path, recovers a strong raw chain at 0.682. This means the original collapse was partly an exposure-bias problem rather than proof that raw sparse-code chaining is impossible. Re-grounding remains the cleanest and strongest stabilization mechanism in the artifact record, but it is no longer the only viable one.

Table 2: Spliced top-1 at 2% retained with global masking.

Section	Field	Vector	Gain
Q1	0.714	0.638	+7.6 pp
Q2	0.716	0.633	+8.3 pp
Q3	0.708	0.665	+4.3 pp
Q4	0.711	0.710	+0.1 pp
Q5	0.715	0.712	+0.3 pp

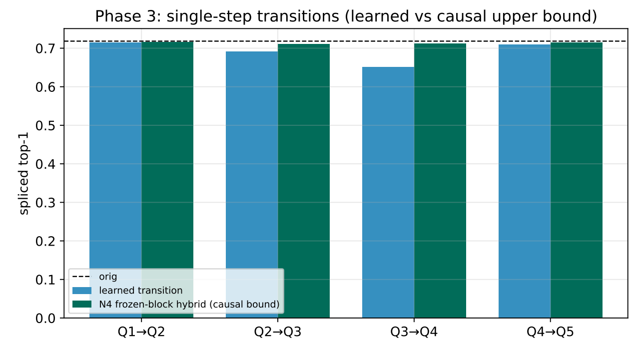

Figure 4: Single-step transition results. The learned $Q1 \rightarrow Q2$ transition matches the frozen-block hybrid upper bound, while $Q3 \rightarrow Q4$ is the consistent bottleneck.

6 Intervention, Sparsity, and Worked Examples

Beyond top-line accuracy, the project also asks whether the sparse codes are meaningful and structured.

Intervention runs show that removing top-magnitude parent coefficients damages downstream behavior 3.3–7.5× more than ablating random kept coefficients, and 38–50% of child-code change concentrates in the parent's receptive-field neighborhood. These results upgrade the hierarchy claim from purely predictive to causal and spatially directed. Feature audits additionally show low dead-feature and duplicate rates, indicating that the learned dictionaries are healthy rather than collapsed.

7 Robustness and Transfer

Phase 4 asks whether the winner-path story survives scale, seed variation, architecture shift, and a first transformer transfer.

These follow-ups materially strengthen the paper's empirical position. The winner-path conclusions are not single-seed accidents, not specific to one CNN depth profile, and not obviously tied to only one architecture family. The transition taxonomy result is especially important: sparse is not merely a reconstructive shortcut, but also the most behavior-preserving transition carrier among the tested alternatives.

Table 3: Single-step transition accuracy versus the frozen-block hybrid upper bound.

Transition	Learned	Hybrid
Q1 → Q2	0.715	0.715
Q2 → Q3	0.691	0.711
Q3 → Q4	0.650	0.711
Q4 → Q5	0.709	0.715

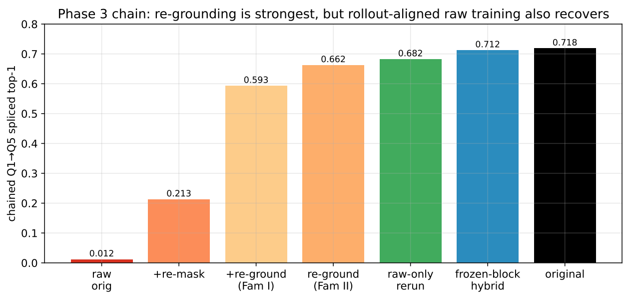

Figure 5: Chained rollout variants. The hybrid chain nearly preserves original accuracy, while naive raw chaining collapses and re-grounding recovers much of the gap.

8 Limitations

The project also has clear boundaries:

- The development center of gravity is still one main anchor setting: ResNet-56 on CIFAR-100.
- The original Family II matrix breadth was not completed, even though chain-aware training on the winner path is substantial.
- The Q3 → Q4 transition remains the hardest step and is not fully solved by scaling predictor capacity alone.
- Patchwise changed reruns are informative but partially incomplete, especially for the depth-aware sweep.
- The ViT result is currently a strong transfer artifact rather than a full task-accuracy paper story.

These limitations narrow the strongest defensible claim: sparse field codes are clearly sufficient and behavior-preserving on the executed winner paths, and hierarchical rollout is real but sensitive to training distribution and manifold consistency.

9 Conclusion

The overall empirical picture is coherent. CNN activation fields are highly compressible in a learned sparse transform domain, and a convolutional Field SAE preserves classifier behavior while retaining only 2–5% of

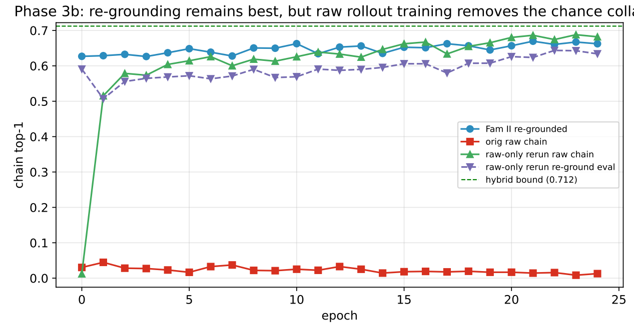

Figure 6: Chain-aware training improves the re-grounded learned chain from 0.593 to 0.662, narrowing the gap to the 0.712 causal upper bound.

Table 4: End-to-end chain accuracy from $Q1$ to $Q5$.

Variant	Top-1
Raw learned chain	0.012
Re-mask each step	0.213
Re-ground each step	0.593
Scheduled-sampling + re-ground	0.662
BPTT through re-grounding	0.701
Frozen-block hybrid chain	0.712
Original classifier	0.718

feature-location coefficients. Sparse codes also carry real hierarchical information: they support strong one-step prediction, and their support is causally sufficient to drive the true network almost end-to-end at original performance. The main challenge is not whether sparse hierarchy exists, but how to train learned chains so that rollout stays on-manifold. Re-grounding, chain-aware training, and rollout-aligned retraining all help, with the best learned chain reaching 0.701 against a 0.712 hybrid upper bound and a 0.718 original baseline.

Taken together, the results support a practical and scientific claim: vision backbones compute with a sparse, spatially structured intermediate representation that can be compressed, propagated, and probed without destroying behavior. That makes sparse field codes a promising substrate for future mechanistic work, especially causal parent-child interventions, improved handling of the Q3 → Q4 bottleneck, and broader extension to transformer architectures.

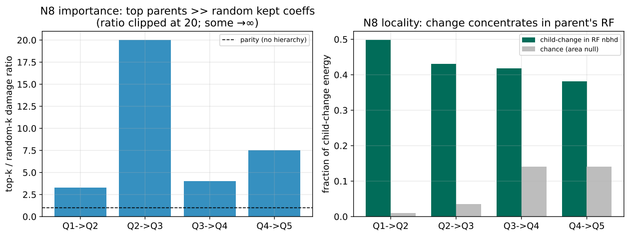

Figure 7: Parent-to-child intervention evidence. Ablating high-importance parent coefficients causes much larger downstream damage than random ablation, and changes concentrate locally in receptive-field neighborhoods.

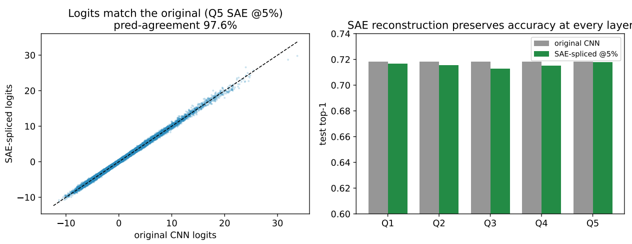

Figure 8: Worked comparison between SAE-spliced and original CNN behavior. Predictions agree on 97.6% of test examples in the shown setting.

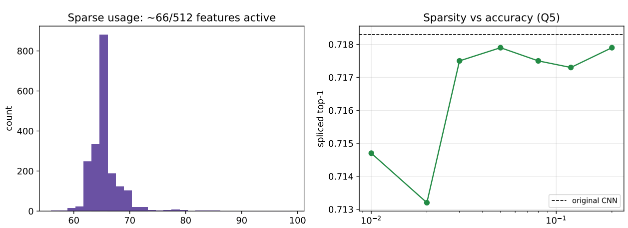

Figure 9: Explicit sparsity diagnostics. Accuracy stays nearly flat down to a small retained fraction, then degrades sharply once the budget becomes too small.

Table 5: Selected Phase 4 extension results.

Experiment	Main finding
ResNet-20 / CIFAR-10	All 15 cells succeed; deeper sections saturate near 92.3% by 5% retained.
Seed stability	Three seeds on ResNet-56/CIFAR-100 remain tightly clustered; means span 0.713–0.718.
Transition taxonomy	Sparse is the best $Q4 \to Q5$ carrier: 0.706 vs 0.690 dense and 0.610 cross.
ResNet-110	The 5% Field SAE preserves 0.723–0.732 top-1 across sections.
ViT transfer	At block 6 and 5% retained, prediction agreement is 94.4% with KL 0.043.

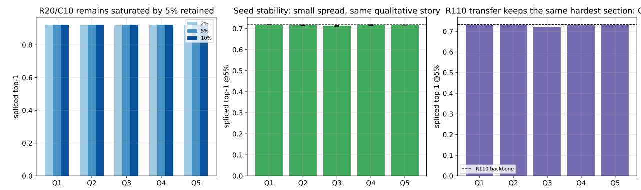

Figure 10: Phase 4 robustness summary. Winner-path behavior remains stable under seed variation and on a new ResNet-20/CIFAR-10 setting.

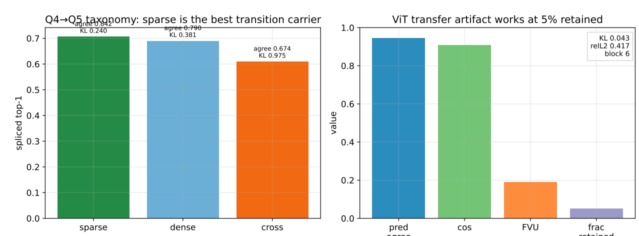

Figure 11: Phase 4 transfer summary. Sparse reconstruction extends to deeper CNNs and yields a first working ViT artifact.